

The Extension Problem for Orthomodular Observation Algebras

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1 Background

All measurement is finite. An experimenter performs finitely many trials, records finitely many digits, distinguishes finitely many outcomes. Yet the mathematical frameworks we use to organise empirical knowledge — probability theory, quantum mechanics, statistical mechanics — rest on infinite structures: σ -algebras, Hilbert spaces, continuous symmetry groups. The passage from finite observation to infinite structure is not automatic; it requires a commitment that no finite body of evidence can compel.

In probability theory, the locus of this commitment is *countable additivity*. Given a Boolean algebra $\mathcal{E}$ of events and a finitely additive charge $\ell : \mathcal{E} \rightarrow [0, 1]$ with $\ell(\Omega) = 1$, the condition

$$E_n \downarrow \emptyset \implies \ell(E_n) \rightarrow 0 \quad (1)$$

— that mass cannot persist along a vanishing sequence of events — is not derivable from any finite collection of observations. De Finetti [9] argued that only finite additivity is empirically grounded; Kolmogorov [22] adopted σ -additivity as an axiom and built modern probability on it. The question of *when* and *why* the passage from one to the other is justified has never been fully resolved.

In quantum theory, the same tension appears in a different guise. The algebra of propositions about a quantum system is not Boolean: incompatible observables cannot be simultaneously determined. The closed subspaces of a Hilbert space H form an orthomodular lattice $L(H)$ — a structure in which complementation is well-defined but distribution of meet over join may fail:

$$a \wedge (b \vee c) \neq (a \wedge b) \vee (a \wedge c) \quad \text{in general.} \quad (2)$$

A *state* on such a lattice is a function $s : L(H) \rightarrow [0, 1]$ satisfying

$$s(\mathbf{1}) = 1, \quad s(a \vee b) = s(a) + s(b) \quad \text{whenever } a \perp b. \quad (3)$$

The non-distributivity means that s is additive only on orthogonal pairs, not on arbitrary finite unions. The classical machinery for extending finitely additive assignments to σ -additive ones — Carathéodory's outer-measure construction, which requires a Boolean algebra of sets — does not apply.

For the specific lattice $L(H)$, this difficulty is resolved by a remarkable rigidity. Gleason's theorem [17] shows that, when $\dim H \geq 3$, every state has the form

$$s(P) = \text{tr}(\rho P) \quad (4)$$

for a unique density operator ρ , and is therefore automatically σ -additive *on the lattice*. The geometry of Hilbert space is so constraining that, for additivity on $L(H)$ itself, the problem of the infinite does not arise — states have no room to misbehave. (On the *extension* axis —

passage to a measure on the dual $S_0(L(H))$ — even these rigid states fail, by §2.1; the rigidity is descent-side only.)

But Hilbert space is very special. Orthomodular lattices arise in many other settings — the projection lattices of operator algebras, the logics of quantum systems with superselection rules, and the abstract lattices studied in quantum logic — and for these, no analogue of Gleason’s rigidity is known. The question that probability theory and quantum theory share — when does finite coherence force countable behaviour? — remains open in this broader setting, and is the subject of the present note.

The programme “Structure from Observation” [24] approaches this question from a particular direction: rather than beginning with a pre-given space Ω of outcomes and asking which σ -algebras it supports, it begins with the *observations themselves* — a directed system of event algebras $\mathcal{E}_i$ with compatible charges ℓ_i — and asks what the directed structure alone determines.

Paper I answers this for Boolean event algebras. A compatible family of finitely additive charges extends to a σ -additive probability measure if and only if collective exhaustion (CE) holds — condition (1) applied uniformly across the directed system. Two constructions realise the extension:

- **Carathéodory:** assumes CE; produces a measure on $(\Omega, \sigma(\mathcal{C}))$ directly.
- **Stone:** assumes nothing; produces a Baire measure on the Stone space $\text{St}(\mathcal{C})$ unconditionally.

Under discriminability, the two agree. Three questions orient the broader programme:

- (1) What must be added to observation to get structure?
- (2) Reconstruction without points: what structure is already present in the relational description — contexts, directed refinement — before point-spaces are assumed?
- (3) Local-to-global extension: what conditions on the *site* control whether local data extend globally?

Paper I resolves (2) and (3) for Boolean algebras. The present note asks whether any of this survives when the event algebra is orthomodular but not distributive.

Epperson and Zafiris [14] advocate a *relational realist* ontology in which structure is constituted by relations among contexts, not by points in a pre-given space. An orthomodular lattice A is the natural algebraic expression of this: its elements are propositions, its partial order is entailment, and its non-distributivity reflects the impossibility of assigning simultaneous truth values to all propositions. The dual space $S_0(A)$ of McDonald–Bimbó [25] is built from *filters* — consistent collections of propositions — not from points of a pre-existing space. This is question (2) applied to non-Boolean algebras: what structure does the relational description already contain?

In the Boolean case, one might hope that the passage from finite to countable additivity could be understood as a *sheaf condition* — local states gluing to a global σ -additive measure. Biesel [1] shows this hope is disappointed:

- Finitely additive charges form a sheaf for finite covers.
- σ -additive measures form a sheaf for countable covers.
- The step between is the *definition* of σ -additivity, not a structural theorem.

This is the boundary at which the present problem begins.

Paper II [23] introduces three positions on the relationship between observation and reality, each a constraint on where the dual-space measure lives:

Empirical adequacy (EA)

The dual-space measure agrees with all observations.

Probabilistic realism (PR)

The measure descends to a realisation space.

Value-definite realism (VDR)

A global two-valued homomorphism exists.

These are nested: $VDR \Rightarrow PR \Rightarrow EA$. For Boolean algebras, all three are commensurable — every EA-theory admits a VDR-completion. For OMLs with $\dim \geq 3$, Gleason’s theorem (4) secures a σ -additive measure on the *lattice* $L(H)$, while Kochen–Specker [21] blocks VDR. The extension problem sits at the $EA \rightarrow PR$ transition: can we pass from a state on the algebra (EA) to a σ -additive measure on the dual space (PR)? Paper II defines PR as a measure on an underlying state space obtained by *descent from the dual-space measure*, so PR presupposes the dual-space measure exists. §2.1 shows that for $L(H)$ this presupposition *fails*: no normal state yields a charge on $S_0(L(H))$, so the dual-space measure does not exist and PR cannot be reached — the chain breaks at the $EA \rightarrow PR$ extension step, not at descent. This **contradicts Paper II’s Commensurability theorem (b)**, which attributes the $L(H)$ dual-space measure to Gleason; §2.1 shows Gleason gives only the *lattice* measure, not the dual-space one. The $L(H)$ verdict flips from “PR holds (Gleason)” to “PR fails for normal states” — a correction to Paper II pending the author’s decision (see the session record), not resolved here. Under what conditions, then — and is descent a free choice (as in the Boolean case) or a structural consequence of the algebra and state together?

Van Fraassen’s distinction [33] between empirical adequacy and structural commitment, which Paper I makes precise via the Stone construction, thus acquires a sharper form in the OML setting: the Stone route is no longer available, the Carathéodory route is blocked by non-distributivity, and the passage from coherent local data to global probability — question (3) — requires genuinely new ideas.

2 The problem

Let A be an orthomodular lattice (OML) and let $s : A \rightarrow [0, 1]$ be a state (3). The McDonald–Bimbó duality [25] associates to A a compact topological space

$$S_0(A) = (F(A), \subseteq, \perp_A, P(A), T(S)),$$

where $F(A)$ is the set of filters on A and $P(A) \subseteq F(A)$ is the set of prime filters. For each $a \in A$, set

$$h(a) = \{x \in F(A) : a \in x\}.$$

The map h is an OML isomorphism from A onto the $\perp$ -stable clopen sets $\text{CO}(S_0(A))^\dagger$. Define a set function μ on these clopens by

$$\mu(h(a)) = s(a). \tag{5}$$

Question 2.1 (Extension). Under what conditions on the OML A does μ extend to a σ -additive measure on the Baire (or Borel) σ -algebra of $S_0(A)$?

This is the direct OML analogue of the Stone extension theorem established for Boolean algebras in [24].

In the Boolean case, the extension is automatic. The clopens $\text{Clop}(\text{St}(\mathcal{C}))$ form a Boolean algebra, the state s restricts to a finitely additive charge on this algebra, and the Carathéodory extension theorem (or Choksi’s projective-limit argument [7]) produces the σ -additive extension. Compactness of the Stone space gives σ -subadditivity for free.

In the OML case each of these steps breaks:

- (i) $\text{CO}(S_0(A))^\dagger$ is an OML, *not* a Boolean algebra.
- (ii) The state s is additive on orthogonal pairs only, not on arbitrary finite unions.
- (iii) **Carathéodory does not apply** — there is no Boolean algebra of sets on which to run the outer-measure construction.
- (iv) Compactness holds (McDonald–Bimbó, Lemma 3.5 [25]), but the σ -subadditivity argument requires orthogonality structure that the topology alone does not provide.

The gap is sharp: “finitely additive on a compact Boolean algebra of clopens” automatically extends; “orthogonally additive on a compact OML of clopen $\perp$ -stable sets” does *not*.

For the prototypical case $A = L(H)$ with $\dim H \geq 3$, Gleason’s theorem (4) gives a σ -additive measure on the *lattice* $L(H)$ itself: every state is induced by a density operator. But this lattice-level rigidity does *not* transfer to the *extension* axis — the passage to a charge on the dual $S_0(L(H))$ — which §2.1 shows fails for every normal state. (The two-axis split below is exactly what separates “measure on $L(H)$ ” from “measure on $S_0(L(H))$.”) And since extension fails, no dual measure exists to descend, so the descent question does not arise for normal states on $L(H)$. For general OMLs, no Gleason-type representation is available even at the lattice level. Question 2.1 asks for the structural conditions that substitute for the Hilbert-space geometry.

Feature	Boolean (Paper I)	OML (this problem)
Dual space	$\text{St}(A)$ (ultrafilters)	$S_0(A)$ (all filters)
Distinguished subset	$\text{pure}(\Omega)$, not canonical	$P(A)$, canonical
Clopes	Boolean algebra	OML (non-distributive)
Additivity of s	full (finite)	orthogonal pairs only
Extension (to the dual)	automatic (Carathéodory)	fails for normal states (§2.1); Gleason is descent-side
Realisation constrained?	No	Yes (Kochen–Specker [21])

Table 1: Structural comparison of the extension problem.

Two axes: extension and descent. The passage from a state on A to a measure on $S_0(A)$ splits into two axes that the Boolean case fuses but the OML case separates. Since $S_0(A)$ is a Stone space, its *full* clopen algebra $\text{CO}(S_0(A))$ is Boolean, whereas the state lives only on the $\perp$ -stable clopens $\text{CO}(S_0(A))^\dagger$ — the OML, the image of h . These share only meet; join, complement, and bottom differ.

Extension Extend μ from the $\perp$ -stable clopens to a finitely additive charge on the *full* Boolean algebra of clopens — the classical Boolean extension problem [19]. Since h preserves meets, $a \wedge b = 0$ forces $h(a) \cap h(b) = h(0)$, so meet-zero elements of A map to *disjoint* clopens. A charge must therefore satisfy, for every pairwise-meet-zero family $\{a_i\}$, $\sum_i s(a_i) \leq 1$ — a Pitowsky correlation polytope / Bell–Boole inequality system [27]. σ -additivity of the state does not help here.

Compactness

Once extension succeeds, the charge extends to a σ -additive Borel measure automatically. Unconditional.

In the Boolean case (Paper I), the extension axis is *free*: finite additivity on the clopen Boolean algebra plus compactness gives the Stone measure unconditionally — “the first arrow is free.” In the OML case, **the extension axis is no longer free.**

The extension obstruction is meet-zero versus orthogonal. Three strengthening conditions on s must be distinguished (Pták–Pulmannová [30], Def. 2):

- (1) *State*: additive on *orthogonal* pairs ($a \leq b^\perp$).
- (2) *Valuation*: additive on *meet-zero* pairs ($a \wedge b = 0$) — a binary condition, strictly stronger, since in a non-distributive OML meet-zero is coarser than orthogonality.
- (3) *Extends to a Boolean charge*: the n -ary condition $\sum_i s(a_i) \leq 1$ for every pairwise-meet-zero family — strictly stronger again.

The extension axis is condition (3). It can fail for *every* state. On the smallest non-distributive OML MO_3 (three 2×2 blocks pasted at 0 and 1), all six atoms are pairwise meet-zero, so orthoadditivity forces the three complementary pairs to sum to 3 while a charge demands ≤ 1 : no state extends. The maximally mixed state $s \equiv \frac{1}{2}$ *is* a valuation (it satisfies the binary condition (2)) yet already fails the n -ary condition (3) at the triple $\{p, q, r\}$, where $\frac{1}{2} + \frac{1}{2} + \frac{1}{2} = \frac{3}{2} > 1$. So the obstruction is the gap between meet-zero and orthogonality, not σ -additivity and not non-contextuality.

The concrete/non-concrete boundary is not where the gap closes. A set-representable OMP (P, L) requires only closure under complement and under *disjoint* union (Burešová–Pták [4], Def. 1.1); closure under intersection is *not* required — it would force L to be a field of sets, hence Boolean. The lattice meet $a \wedge b$ is the greatest L -member contained in $A \cap B$, so $a \wedge b \subseteq A \cap B$ with equality iff $A \cap B \in L$. Hence orthogonality ($A \cap B = \emptyset$) implies meet-zero always, but **meet-zero does not imply orthogonality** for incompatible pairs. The smallest witness is MO_2 , the concrete logic on $P = \{1, 2, 3, 4\}$ with $L = \{\emptyset, \{1, 2\}, \{3, 4\}, \{1, 3\}, \{2, 4\}, P\}$: there $\{1, 2\} \wedge \{1, 3\} = 0$ while $\{1, 2\} \cap \{1, 3\} = \{1\} \neq \emptyset$. Crucially, MO_3 *itself is concrete* (it carries 2^3 ordering two-valued states, hence is set-representable by Gudder’s representation theorem for concrete logics — a concrete logic is exactly an OMP with an order-determining set of two-valued states [29]), so the flagship “meet-zero $\neq$ orthogonal” example above is a concrete logic. The boundary that matters is therefore not concrete versus non-concrete but a *richness* / atomicity condition: meet-zero coincides with orthogonality iff every nonempty $A \cap B$ contains a nonzero member of L (sufficient: all singletons lie in L). This property has no established standard name (its cousins — Jauch–Piron, Tkadlec “regional,” Burešová–Pták “point-distinguishing” — are all distinct). (Terminology trap: MO_3 ’s celebrated non-representability is von Neumann *coordinatisation* — failure to be a subspace lattice of a projective geometry — a different notion from set-representability.)

Status of the extension axis. For *rich* concrete OMLs (singletons in L , or the weaker intersection-richness above) meet-zero coincides with orthogonality, and the extension axis reduces to the classical marginal / Pitowsky non-contextuality problem. For *non-rich* concrete OMLs (MO_2 , MO_3) and non-concrete OMLs ($L(H)$) the meet-zero/orthogonal gap is non-empty, and the extension axis is strictly stronger and can fail universally. In the *finite* case it is a decidable linear program, not a structurally resistant frontier. The *infinite* case — $L(H)$, to which the finite counterexample does not directly speak — is resolved for normal states by the clustering argument of §2.1 below, with only the singular states left open.

2.1 The infinite extension case for $L(H)$

For $L(H)$ with $\dim H = \infty$ the extension axis admits a decisive finite-dimensional argument, sharper than the MO_3 linear program. Fix a 2-dimensional subspace $e \subset H$. For any line $p \subset e$,

its perpendicular $p^\perp \subset e$ satisfies $p \perp p^\perp$ and $p \vee p^\perp = e$, so orthoadditivity gives

$$s(p) + s(p^\perp) = s(e). \quad (6)$$

Choose k distinct lines $p_1, \dots, p_k \subset e$ in general position, so that the $2k$ lines $\{p_i, p_i^\perp\}$ are pairwise distinct. Distinct lines meet only at 0, hence are pairwise meet-zero, so h sends them to $2k$ pairwise-disjoint clopens. A finitely additive charge extending $\mu(h(a)) = s(a)$ must therefore satisfy, using (6),

$$\sum_{i=1}^k [s(p_i) + s(p_i^\perp)] = k s(e) \leq 1 \quad \text{for every } k \geq 1. \quad (7)$$

If $s(e) > 0$, taking $k > 1/s(e)$ violates (7). Hence:

Proposition 2.2. *Any state s on $L(H)$ with $s(e) > 0$ for even one finite-dimensional e fails to extend to a charge on the full clopen algebra of $S_0(L(H))$. In particular no normal (Gleason) state extends. The argument uses only orthoadditivity on lines in a single 2-plane; σ -completeness of the lattice is irrelevant.*

Every normal state $s = \text{tr}(\rho \cdot)$ has $s(e) > 0$ for some 2-dim e (as $\rho \neq 0$), so Proposition 2.2 kills the entire σ -additive (Gleason) state space. This answers the infinite extension question for the physically central states, and answers “does σ -completeness change it?” in the negative.

The only states the argument does *not* reach are the *singular* ones: $s(e) = 0$ for every finite-dimensional e , where the bound (7) is vacuous. These exist (take H separable) — e.g. the ultrafilter vector state $s(a) = \lim_\omega \langle a e_n, e_n \rangle$ for a free ultrafilter ω and an orthonormal basis (e_n) , or any pullback of a state on the Calkin algebra $B(H)/K(H)$. Moreover, lifting a lattice state to a bounded functional φ on $B(H)$ by Mackey–Gleason / Bunce–Wright [3, 8] (valid as $B(H)$ has no type I_2 summand), the Takesaki normal/singular decomposition $\varphi = \varphi_n + \varphi_s$ with both parts positive [32] writes $B(H)^* = (\text{trace class}) \oplus K(H)^\perp$ — where the singular part φ_s *annihilates the compacts* $K(H)$ (a statement at the level of φ , after the lift, not of the bare lattice state). A finite-rank projection e then has $s(e) = \text{tr}(\rho e)$, and $s(e) = 0$ for all finite-dim e forces $\rho = 0$ (taking e rank-one gives $\langle \rho \xi, \xi \rangle = 0$ for all ξ). So a state is *either* detected by some finite-rank projection (killed by Proposition 2.2) *or* purely singular — a clean dichotomy with no third class (the decomposition being unique). The density $\varphi_n = \text{tr}(\rho \cdot)$ is named by predual duality $B(H)_* = \text{trace class}$, *not* by Gleason. Whether a singular orthoadditive state extends is the open residue; there the relevant meet-zero families are infinite-dimensional subspaces with trivial intersection, and σ -additivity *of the state* (not of the lattice) is what bites. (Proposition 2.2 is the proof; the inequality is instance-checked — finitely many lines in $\mathbb{R}^2$, the charge LP infeasible once $k s(e) > 1$ — by `verification/lh_infinite_extension.py`, which does not itself touch the infinite-dimensional $S_0(L(H))$). The singular/normal dichotomy is hand-verified — foundations cited with hypotheses checked, gluing checked by hand — in `verification/lh_singular_dichotomy.md`.)

Descent. σ -additivity of the state governs descent, not extension. Even once μ extends to a measure $\hat{\mu}$ on $S_0(A)$, the *descent* question — whether $\hat{\mu}$ concentrates on the “physical” points — changes character:

- | | |
|----------------|--|
| Boolean | Does $\hat{\mu}$ concentrate on $\text{pure}(\Omega)$? If and only if the charges are σ -additive. The choice of realisation space Ω is unconstrained. |
| OML | Does $\hat{\mu}$ concentrate on $P(A)$? Here $P(A)$ is part of the structure, not a free choice. The Kochen–Specker obstruction [21] prevents simultaneous realisation of all filters; the Bub–Clifton theorem [2] shows that the state determines a maximal Boolean subalgebra of definite values. |

In the OML setting, the algebra and state *together* constrain which filters are “realised.” Descent is not a geometric commitment but a structural consequence.

Remark 2.3 (Decomposition lives on the descent axis). De Simone–Navara [10] decompose OMP states as $s = s_\sigma + s_{\text{wpfa}}$ (a σ -additive part and a weakly purely finitely additive part), the orthomodular analogue of Yosida–Hewitt. By the axis split above, the wpfa component governs the *descent* axis — it is the analogue of the purely finitely additive part ℓ_p in the Boolean case — and *not* the extension axis. A natural-looking conjecture, “ s extends iff $s_{\text{wpfa}} = 0$,” is therefore mis-targeted: vanishing of s_{wpfa} concerns whether the (already σ -additive) state’s measure concentrates on $P(A)$, while extension is blocked by non-distributivity regardless of the wpfa component. The decomposition theory and the extension problem live on different axes.

A caveat compounds this. The McDonald–Bimbó duality [25] is *finitary*: the identity $h(\bigvee_n a_n) = (\bigcup_n h(a_n))^{\perp\perp}$ holds for finite joins but fails in general for countable joins, since finitary OML homomorphisms need not preserve infinite suprema. Making even the descent argument rigorous — the analogue of Paper I’s “ σ -additivity $\Leftrightarrow$ concentration” — requires σ -completeness and continuity hypotheses the 2023 duality does not supply, and is itself open.

The finitary-to- σ bridge: open, with three named routes blocked. Paper I’s descent runs on a *Loomis–Sikorski* engine: σ -additive measures on the Stone space correspond to σ -homomorphisms off the ideal points, which is what makes “ σ -additive $\Leftrightarrow$ concentrates on pure(Ω)” go through. The OML descent question is whether that engine has an OML analogue — whether a σ -complete OML admits a Loomis–Sikorski representation (quotient of a σ -tribe of sets by a σ -ideal) or a countable-join-preserving σ -Stone duality. The literature settles only the known *routes*, all of which are blocked.

- (i) **RDP / effect-algebra route — blocked.** Loomis–Sikorski holds for σ -complete MV-algebras (Dvurečenskij [12]) and for monotone σ -complete effect algebras *with* the Riesz Decomposition Property (RDP) [13]. But a lattice effect algebra has RDP iff it is an MV-algebra (Riečanová; Paseka), and an OML that is an MV-algebra — all pairs compatible — is Boolean (Kalmbach [20]). So $\text{OML} + \text{RDP} \Leftrightarrow \text{Boolean}$: the RDP machinery has no non-Boolean OML to act on.
- (ii) **MacNeille-completion route — blocked.** Harding [18] shows the MacNeille completion of an OML need not be orthomodular, so one cannot complete to absorb countable joins.
- (iii) **σ -Stone-duality route — none exists.** McDonald–Bimbó is irreducibly finitary; Freytes’ equational theory of σ -complete OMLs [16] supplies no set/tribe representation.

A concentration statement needs a point space, which a general OML may lack (Greechie stateless lattices are non-concrete). But that is a red herring for the live case: $L(H)$ is non-concrete yet Gleason hands it a clean lattice measure (the non-concreteness is no obstacle there), and MO_3 is concrete. The genuinely open class is **concrete, non-Boolean, infinite σ -OMLs**, where the non-concreteness obstruction is absent and the RDP argument blocks only one route. No impossibility theorem is known for this class; the contribution here is the map of dead routes, which sharpens “needs a new idea” into “needs a representation bypassing all three.”

Two grades of probabilistic realism. The two axes, re-expressed in the EA/PR/VDR vocabulary of Paper II [23], split *probabilistic realism* (PR) into two grades that the Boolean case fuses and non-distributivity separates — the same extension/descent split, named in the realism language, not a new construction.

PR_{lat} A σ -additive measure on the *lattice* A agreeing with s . For $L(H)$, $\dim H \geq 3$, Gleason supplies this for every σ -additive state.

PR_{dual} A σ -additive Borel measure $\hat{\mu}$ on the *dual* $S_0(A)$ with $\hat{\mu}(h(a)) = s(a)$ concentrating on the physical points, $\hat{\mu}(P(A)) = 1$ — extension (§2) followed by descent.

The relation between the two grades is a *witnessed one-directional separation*, not a proven nesting. $L(H)$ normal states *separate* them: they clear PR_{lat} (Gleason) and fail PR_{dual} (§2.1: no charge on $S_0(L(H))$ exists, a fortiori none concentrating). The reverse inclusion $\text{PR}_{\text{dual}} \Rightarrow \text{PR}_{\text{lat}}$ is *not* free: restricting a dual measure $\hat{\mu}$ to the $\perp$ -stable clopens recovers s only as a *finitely* orthoadditive state, whereas PR_{lat} needs σ -additivity on the lattice, $s(\bigvee_n a_n) = \sum_n s(a_n)$. The gap is $\hat{\mu}(D)$, $D = h(\bigvee_n a_n) \setminus \bigcup_n h(a_n)$, and concentration on $P(A)$ does *not* force $\hat{\mu}(D) = 0$: D contains *principal* filters (e.g. $\uparrow p$ with $p = \text{span}(v)$, v of infinite support, $a_n = \text{span}(e_n)$: $p \leq \bigvee_n a_n$ but $p \not\leq a_n$ for all n). So $\text{PR}_{\text{dual}} \Rightarrow \text{PR}_{\text{lat}}$ requires an extra σ -continuity hypothesis ($\hat{\mu}$ null on countable-join defects) that the finitary McDonald–Bimbó duality does not supply — the *same* finitary wall flagged above (verified against the primary source, `verification/mb_primeness_check.md`). What *is* established is the separation, and it lets one assert *both* grades at once — Gleason clears PR_{lat} for $L(H)$ while PR_{dual} fails — without suppressing either. In the Boolean case the two grades coincide (Stone gives $\hat{\mu}$ unconditionally; descent $\Leftrightarrow \sigma$ -additivity), which is why Paper II could write a single PR. The distinction is philosophically neutral: a Born-rule realist reads off PR_{lat}, a relational realist PR_{dual}, a constructive empiricist EA.

What is *not* established is that descent does any work independent of extension. Every known PR_{dual} failure is an extension failure; in finite dimensions descent is vacuous (the only non-principal point is the improper filter, already null by $s(0) = 0$ — verified, `verification/pr_dual_inhabita` where PR_{dual} on MO_2 collapses to the extension LP). The descent half can be strictly weaker than extension only in infinite dimensions, where free non-principal filters can carry mass — exactly where the finitary-to- σ bridge above fails. Hence the open question: *is there a σ -OML regime in which descent is both statable and strictly weaker than extension — a non-distributive OML and state that extends to a charge on $S_0(A)$ but fails to concentrate on $P(A)$?* The only candidate is the singular-state sliver of §2.1, which runs into the same statability wall. Until such a witness exists, PR_{lat}/PR_{dual} is a strict stratification with the descent half not yet shown independent — a vocabulary refinement, not a new theorem.

3 What is known

In the Boolean case, the Kelley–Vladimirov–Pták characterisation (Fremlin, Theorem 391D [15]) gives a complete algebraic answer:

Theorem 3.1 (KVP). *A Boolean algebra B carries a strictly positive σ -additive measure if and only if B is*

- (a) *Dedekind σ -complete,*
- (b) *weakly (σ, ∞) -distributive, and*
- (c) *chargeable (carries a strictly positive finitely additive measure).*

Each condition is algebraic and non-trivially constraining. For OMLs, the analogues fail or degenerate:

- (a) Dedekind σ -completeness makes sense but is very strong.
- (b) Weak (σ, ∞) -distributivity has no clean OML analogue — the definition relies on distributivity of $\bigwedge$ over $\bigvee$, which may fail orthomodularly.
- (c) Chargeability has no known structural characterisation on OMLs; the existence of a strictly positive finitely additive measure cannot be read off from the lattice structure.

The positive results that do exist for OMLs all require algebraic structure rich enough to approximate Hilbert-space geometry:

Gleason [17]	$A = L(H)$, $\dim H \geq 3$. Every state has the form (4); σ -additivity <i>on the lattice</i> is automatic (this is lattice-level, not dual-space extension — cf. §2.1).
Bunce–Wright [3]	A the projection lattice of a JBW-algebra. σ -additivity via operator-algebraic structure.
Chetcuti–Dvurečenskij [6]	Lattice effect algebras with completeness conditions close to von Neumann algebras.

The obstructive results show that the Boolean extension machinery cannot be transplanted:

Pták–Pulmannová [29]	If every unital subadditive measure on an orthomodular poset is a state, the poset must be Boolean. This is the structural reason no KVP analogue exists for OMLs: conditions strong enough to force σ -additivity collapse the lattice back to a Boolean algebra.
Navara [26]	Regularity does <i>not</i> force σ -additivity on general orthomodular posets. The Béaver–Cook result is special to von Neumann algebras.
Pták [28]	Exotic OML state spaces exist; the obstruction is combinatorial (Greechie pasting), not from the centre.

Recent incremental work confirms the field is active but no one has found the right condition: Voráček–Pták [34] study signed measures on orthomodular posets; Burešová–Pták [5] investigate variations on regularity conditions; De Simone–Navara study Yosida–Hewitt-type decompositions for orthomodular posets.

For the σ -additivity / descent side, the gap between “too weak” (allows non- σ -additive states) and “too strong” (collapses to Boolean) appears structurally robust — this is the part where further reading has not produced the answer. (This robustness is about the descent axis; the extension axis is classical, per §2.)

All known exotic OML state spaces are built by **Greechie pasting** [35, 31, 28]. Progress on the descent residue likely requires either:

- (a) a new pasting technique that controls σ -additivity, or
- (b) an approach that bypasses the state space entirely (e.g., categorical or topos-theoretic [11]).

4 Assessment

Question 2.1, once split into its two axes, is not a single open problem but a settled axis and an open one.

Extension axis: largely settled, and partly degenerate. This is the classical Boolean extension problem [19] / Pitowsky polytope feasibility [27]. Because h preserves meets, it is governed by the gap between meet-zero and orthogonality, which is non-empty whenever the concrete representation is non-rich (and a fortiori in any non-distributive OML). In the finite case it is a decidable linear program; it can fail for *every* state (MO_3). This is not a structurally resistant frontier and does not require a new idea. Two earlier framings of this axis were mistaken. (i) “The Pták–Pulmannová frontier needing a new idea” — Pták–Pulmannová [30] concerns the *supply* of valuations (a unital set forces Boolean), not whether a given state extends, and extension is in any case strictly stronger than the valuation condition. (ii) “Concrete OMLs reduce the axis to Pitowsky non-contextuality because meet = intersection” — false:

concreteness requires only disjoint-union closure, and MO_2 , MO_3 are concrete logics in which meet-zero $\neq$ orthogonality (§2). The gap closes only under a richness / atomicity hypothesis, not under concreteness.

Descent axis: genuinely open, dead routes mapped. Whether a measure on $S_0(A)$ concentrates on the physical points $P(A)$, and the analogue of Paper I’s “ σ -additivity $\Leftrightarrow$ concentration,” cannot even be stated rigorously without σ -completeness and continuity hypotheses, because the McDonald–Bimbó duality is finitary (the countable-join identity fails). The descent argument runs on a Loomis–Sikorski engine, and the three known routes to an OML analogue are all blocked — the RDP / effect-algebra route ($\text{OML} + \text{RDP} \Leftrightarrow \text{Boolean}$), the MacNeille completion (Harding), and a σ -Stone duality (none exists; Freytes gives only an equational theory). But *no impossibility theorem* is known for the live class — concrete, non-Boolean, infinite σ -OMLs. This is the live residue, now sharpened from “needs a new idea” to “needs a representation bypassing all three blocked routes.”

The infinite extension case: no normal state extends; the singular case is the open sliver. For $L(H)$ with $\dim H = \infty$ the extension fails decisively for every normal state, by a clustering argument inside a single 2-plane (§2.1). Fix a 2-dim subspace e ; for k distinct lines $p_1, \dots, p_k \subset e$, the $2k$ lines $\{p_i, p_i^\perp\}$ are pairwise meet-zero, so a charge forces $\sum_i [s(p_i) + s(p_i^\perp)] = k s(e) \leq 1$ for every k . Hence any state with $s(e) > 0$ for even one finite-dim e fails — which is every normal (Gleason) state. σ -completeness of the lattice is irrelevant; extension is a finitary charge condition. The only escapees are *singular* states ($s(e) = 0$ for all finite-dim e); these exist (e.g. ultrafilter vector states, or pullbacks of Calkin-algebra states) and form, with the normal states, a clean dichotomy (Takesaki normal/singular decomposition via the Bunce–Wright lift [3]). Whether a singular orthoadditive state extends — where the relevant meet-zero families are infinite-dim subspaces with trivial intersection, and σ -additivity of the state is what bites — is the open sliver.

Novelty. The formulation via McDonald–Bimbó duality [25] appears to be new; the identification of the extension axis with classical Horn–Tarski / Pitowsky feasibility de-mystifies it. For $L(H)$, Gleason gives the σ -additive measure on the lattice $L(H)$ itself — but this lattice-level rigidity does *not* transfer to the dual: the clustering argument (§2.1) shows the passage to a charge on $S_0(L(H))$ fails for every normal state. Since extension fails there is no dual measure $\hat{\mu}$ to descend, so the descent question does not even arise for normal states on $L(H)$.

Status. Extension axis: settled (finite; infinite $L(H)$ resolved for normal states by Proposition 2.2, the singular case the open sliver). Descent axis: open, three known routes blocked (RDP / MacNeille / σ -Stone), no impossibility theorem. Not a current active lead.

References

- [1] Owen Biesel. Sheaves of probability. arXiv preprint arXiv:2401.01968, 2024.
- [2] Jeffrey Bub and Rob Clifton. A uniqueness theorem for ‘no collapse’ interpretations of quantum mechanics. *Studies in History and Philosophy of Modern Physics*, 27(2):181–219, 1996.
- [3] Leslie J. Bunce and J. D. Maitland Wright. The Mackey–Gleason problem. *Bulletin of the American Mathematical Society*, 26(2):288–293, 1992.
- [4] Dominika Burešová and Pavel Pták. On the set-representable orthomodular posets that are point-distinguishing. *International Journal of Theoretical Physics*, 62, 2023. Also arXiv:2401.13798.

- [5] Klára Burešová and Pavel Pták. Variations on regularity conditions for orthomodular posets. 2023. Preprint; UNVERIFIED, see note above.
- [6] Emmanuel Chetcuti and Anatolij Dvurečenskij. On the Gleason theorem for effect algebras. *Mathematica Slovaca*, 55(3):297–308, 2005.
- [7] J. R. Choksi. Inverse limits of measure spaces. *Proceedings of the London Mathematical Society*, 8(3):321–342, 1958.
- [8] Erik Christensen. Measures on projections and physical states. *Communications in Mathematical Physics*, 86(4):529–538, 1982.
- [9] Bruno de Finetti. *Theory of Probability*, volume 1. Wiley, 1974.
- [10] Anna De Simone and Mirko Navara. Yosida–Hewitt and Lebesgue decompositions of states on orthomodular posets. *Journal of Mathematical Analysis and Applications*, 255(1):74–104, 2001.
- [11] Andreas Döring and Chris J. Isham. A topos foundation for theories of physics: I. Formal languages for physics. *Journal of Mathematical Physics*, 49(5):053515, 2008.
- [12] Anatolij Dvurečenskij. Loomis–sikorski theorem for σ -complete MV-algebras and ℓ -groups. *Journal of the Australian Mathematical Society*, 68(2):261–277, 2000.
- [13] Anatolij Dvurečenskij. Loomis–sikorski theorem for monotone σ -complete effect algebras. *Journal of the Australian Mathematical Society*, 79(3):305–318, 2005.
- [14] Michael Epperson and Elias Zafiris. *Foundations of Relational Realism: A Topological Approach to Quantum Mechanics and the Philosophy of Nature*. Lexington Books, Lanham, MD, 2013.
- [15] D. H. Fremlin. *Measure Theory, Vol. 3: Measure Algebras*. Torres Fremlin, Colchester, 2004.
- [16] Hector Freytes. An equational theory for σ -complete orthomodular lattices. *Soft Computing*, 24(14):10257–10264, 2020.
- [17] Andrew M. Gleason. Measures on the closed subspaces of a Hilbert space. *Journal of Mathematics and Mechanics*, 6(6):885–893, 1957.
- [18] John Harding. Orthomodular lattices whose MacNeille completions are not orthomodular. *Order*, 8(1):93–103, 1991.
- [19] Alfred Horn and Alfred Tarski. Measures in Boolean algebras. *Transactions of the American Mathematical Society*, 64(3):467–497, 1948.
- [20] Gudrun Kalmbach. *Orthomodular Lattices*. Academic Press, 1983.
- [21] Simon Kochen and Ernst P. Specker. The problem of hidden variables in quantum mechanics. *Journal of Mathematics and Mechanics*, 17(1):59–87, 1967.
- [22] A. N. Kolmogorov. *Foundations of the Theory of Probability*. Chelsea, 1933.
- [23] Patrick S. Mahon. Distributivity and the commensurability of empirical adequacy and realism. Preprint, 2026.
- [24] Patrick S. Mahon. When does observational coherence determine probability? Preprint, 2026.

- [25] J. McDonald and Katalin Bimbó. Duality for orthomodular lattices. 2023. Preprint.
- [26] Mirko Navara. When is the integral on quantum probability spaces additive? *Bulletin of the Polish Academy of Sciences — Mathematics*, 40(1):29–33, 1992.
- [27] Itamar Pitowsky. *Quantum Probability — Quantum Logic*, volume 321 of *Lecture Notes in Physics*. Springer, Berlin, 1989.
- [28] Pavel Pták. Exotic logics. *Colloquium Mathematicum*, 54(1):1–7, 1987.
- [29] Pavel Pták and Sylvia Pulmannová. *Orthomodular Structures as Quantum Logics*, volume 44 of *Fundamental Theories of Physics*. Kluwer, Dordrecht, 1991.
- [30] Pavel Pták and Sylvia Pulmannová. A measure-theoretic characterization of Boolean algebras among orthomodular lattices. *Commentationes Mathematicae Universitatis Carolinae*, 35(1):205–208, 1994.
- [31] Frederic W. Shultz. A characterization of state spaces of orthomodular lattices. *Journal of Combinatorial Theory, Series A*, 17(3):317–328, 1974.
- [32] Masamichi Takesaki. *Theory of Operator Algebras I*. Springer-Verlag, 1979. Ch. III, normal/singular decomposition of the dual.
- [33] Bas C. van Fraassen. *The Scientific Image*. Oxford University Press, 1980.
- [34] Václav Voráček and Pavel Pták. Signed measures on orthomodular posets. 2023. Preprint; UNVERIFIED, see note above.
- [35] Alexander Wilce. Test spaces. In Kurt Engesser, Dov M. Gabbay, and Daniel Lehmann, editors, *Handbook of Quantum Logic and Quantum Structures: Quantum Logic*, pages 443–549. Elsevier, 2009.